

Deep Reinforcement Learning Assignment 1 Documentation

1. Purpose

This repository provides a structured, team-friendly implementation for the assignment divided into

two p

- - **Part 1: Multi-Armed Bandit (MAB)**

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- - **Part 2: Dynamic Programming (DP)**

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2. Problem Statement Summary

MAB Problem

A hospital must recommend treatments for patients with varied disease severity. Each treatment is an

arm v

Key concepts:

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DP Problem

A rescue drone must navigate a disaster grid to save targets, avoid danger zones, manage battery,

and n

Key concepts:

- - C

3. Folder Structure

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File descriptions

- - `README.md` project overview and quick start instructions.

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4. Shared Parameters and Derivation

The shared parameter file is `parameters.py`.

Derived values from group number 91

```
-- `GROUP_NUMBER = 91`
```

```
#### MAB derived parameters
```

```
-- `MAB_NUM_MEDICINES = (GROUP_NUMBER % 3) + 5 = 6`
```

```
#### DP derived parameters
```

Since the last digit is `1`, the DP settings are:

```
#### Feature toggles
```

```
-- `MAB_ENABLED = 1`
```

5. How to Change Parameters

1. Open `parameters.py`.

If you want to enable or disable a part of the assignment, set:

- - >

6. How to Run the Flow

Setup

Install dependencies once:

```
```bash
```

pip in

### Run MAB workflow

Open and execute ``mab/run_mab.ipynb`` in Jupyter Notebook or JupyterLab.

### Run DP workflow

Open and execute ``dp/run_dp.ipynb`` in Jupyter Notebook or JupyterLab.

### Optional: run modules from Python

You can import functions directly from the modules in the scripts.

## 7. Purpose of MAB and DP in this Assignment

## MAB Purpose

- - Demonstrate learning from sequential clinical trial data.

## DP Purpose

- - Demonstrate modeling an MDP for robotic decision-making.

## 8. Making This a Team Effort

### Suggested team roles

- - **Member 1:** MAB dataset and environment setup.

### How to collaborate

- - Use the notebook ``run_mab.ipynb`` and ``run_dp.ipynb`` as working documents.

## 9. Update Process

To update the parameter file and propagate changes:

1. Modify `GROUP\_NUMBER` in `parameters.py`.

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For future extensions, add new modules under `mab/` or `dp/` and reference them from the notebooks.

## 10. Notes

- - The framework is designed to be reusable and maintainable.

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